



Introduction to Financial Markets and Derivatives

What is a Derivative?

- A derivative is an instrument whose value depends on, or is derived from, the value of another asset.
- Examples: futures, forwards, swaps, options, exotics...

Commodity derivatives are available on agriculture products, metals other than bullion, bullion, and energy (e.g. light sweet crude oil).

Equity derivatives are futures and options on stocks and indices. For instance, the F&O segment of NSE includes futures and options on 135 stocks and 6 indices.

Currency derivatives are futures and options on different currency pairs. In NSE, futures and option contracts are available for four foreign currencies and INR pairs viz. USD-INR, GBP-INR, EUR-INR, and JPY-INR. In addition, three cross currency futures contracts are available, namely GBP-USD, EUR-USD, and USD/JPY.

Interest rate derivatives are futures contract to buy or sell G-Secs and Treasury Bills. In NSE, interest rate futures are available for G-Secs with maturity period of 6 year, 10 year, 13 year and 91 day T-Bills.

Derivative Markets

- OTC vs. Exchanges
- Cash market (or spot market) – market where underlying is delivered immediately or shortly thereafter



Concepts of Financial and Derivative Markets

- Risk Preference
 - Risk aversion vs. risk neutrality
 - Risk seeker
- Short Selling
- Repurchase agreements (repos)
- Return and Risk
 - Risk defined
 - The risk-return tradeoff

Forward Price

- The forward price for a contract is the delivery price that would be applicable to the contract if were negotiated today (i.e., it is the delivery price that would make the contract worth exactly zero)
- The forward price may be different for contracts of different maturities

Terminology

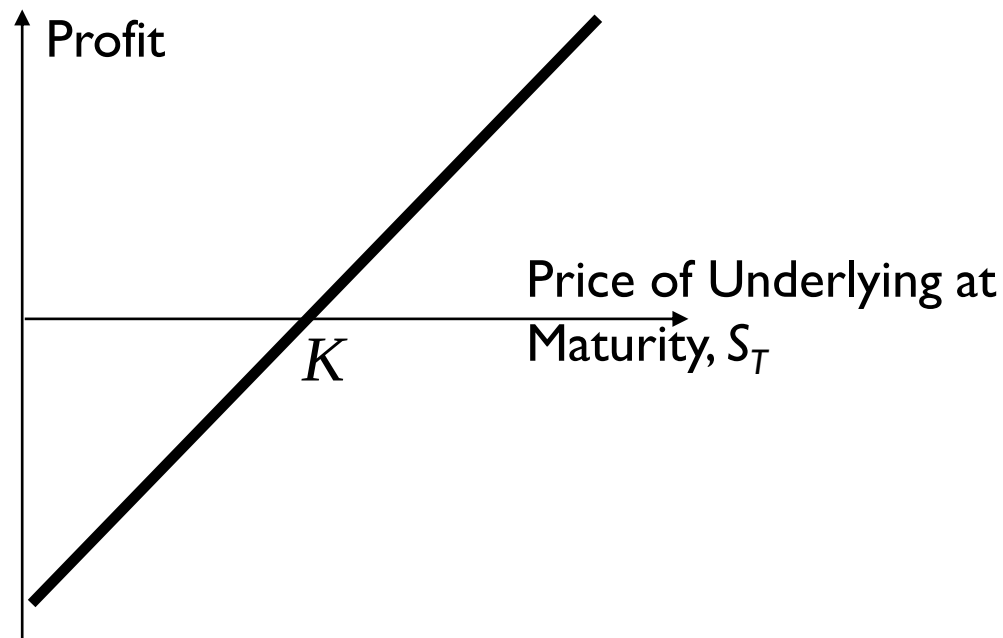
- The party that has agreed to buy has what is termed a long position
- The party that has agreed to sell has what is termed a short position

Example

- On January 5, 2019, a firm enters into a long forward contract to buy GBP of 1 million in six months at an exchange rate of 1.4561 for USD
- This obligates the corporation to pay \$1,456,100 for GBP 1 million on June 5, 2019
- What are the possible outcomes?

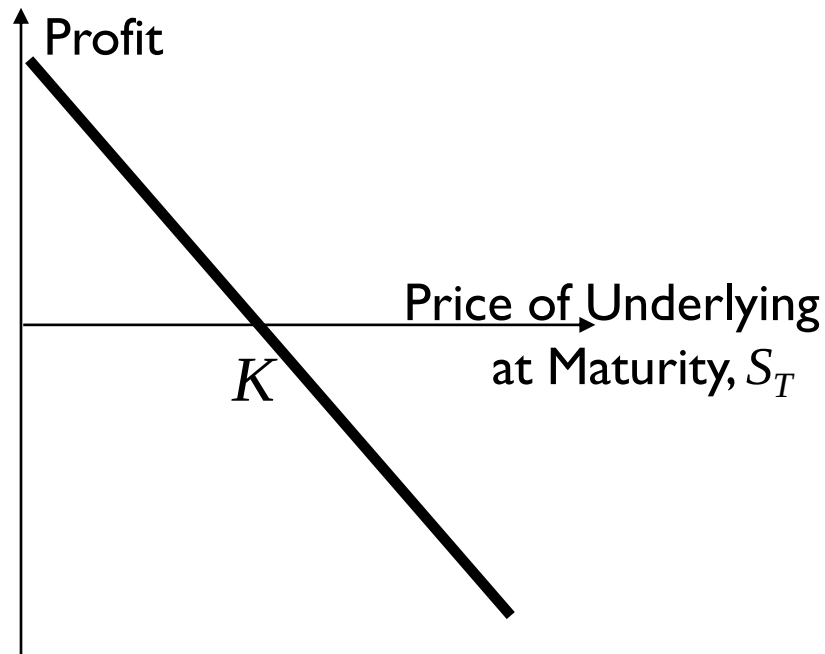
Profit from a Long Forward Position

(K = delivery price = forward price at time contract is entered into)



Profit from a Short Forward Position

(K = delivery price = forward price at time contract is entered into)



Futures Contracts

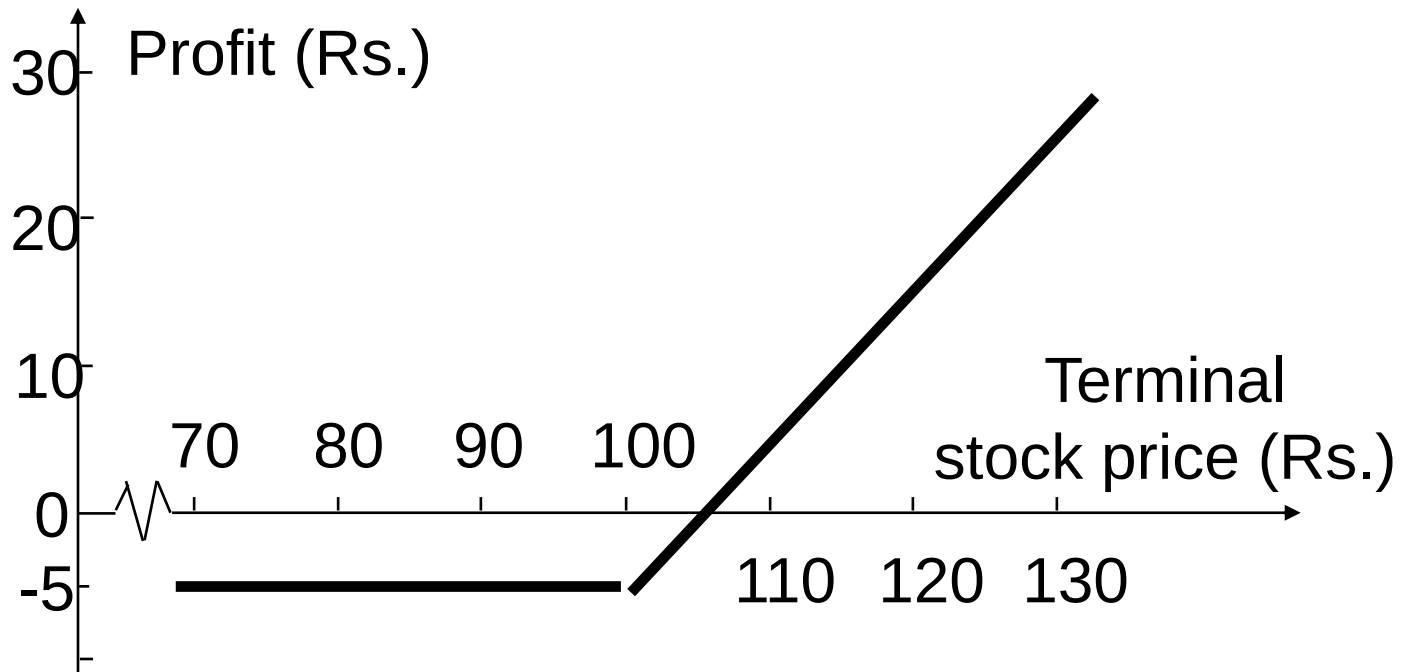
- Agreement to buy or sell an asset for a certain price at a certain time
- Similar to forward contract
- Whereas a forward contract is traded OTC, a futures contract is traded on an exchange

Options

- Premium – price paid to buy an option
- Call – right to buy
- Put – right to sell
- Exercise or strike price – fixed price to option buyer and buy (call) or sell (put)
- Expiration date – options have finite lives
- European – exercise only on expiration date
- American – exercise anytime before or on expiration date

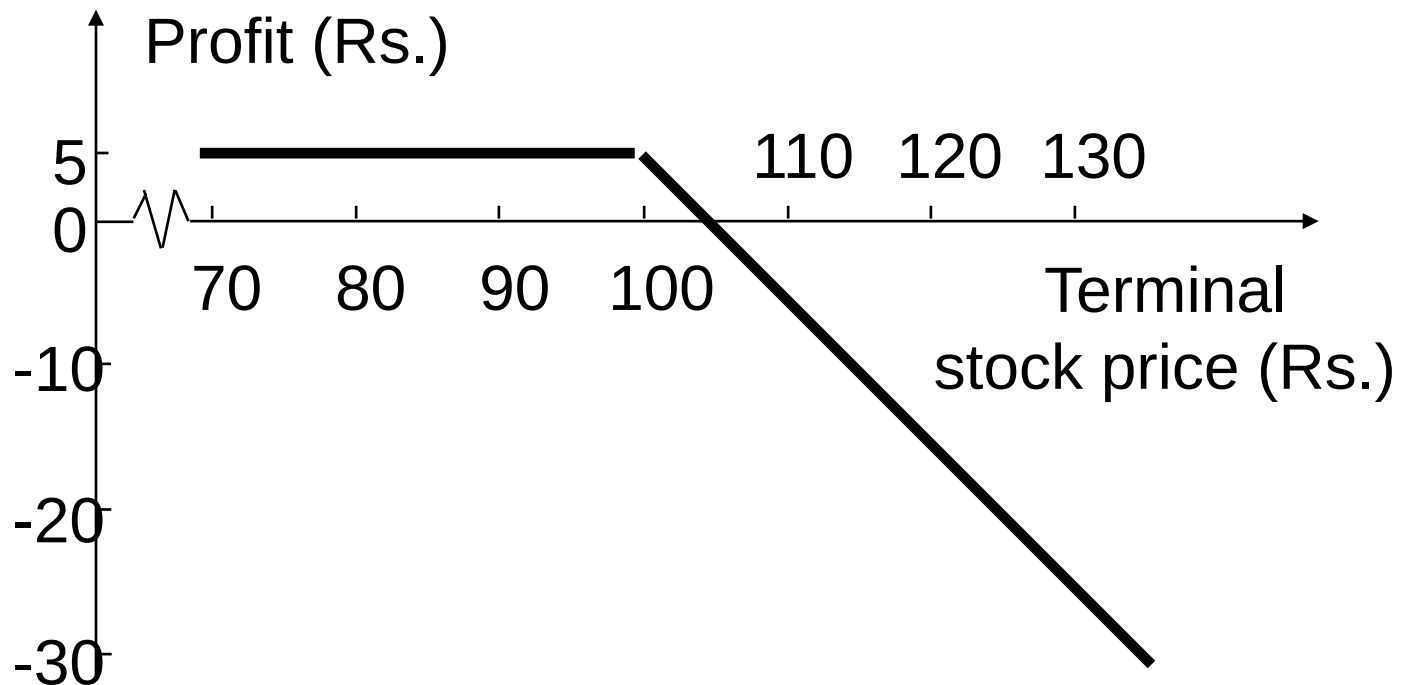
Long Call (Call Holder)

Profit from buying one European call option: option price = Rs. 5, strike price = Rs. 100, option life = 2 months



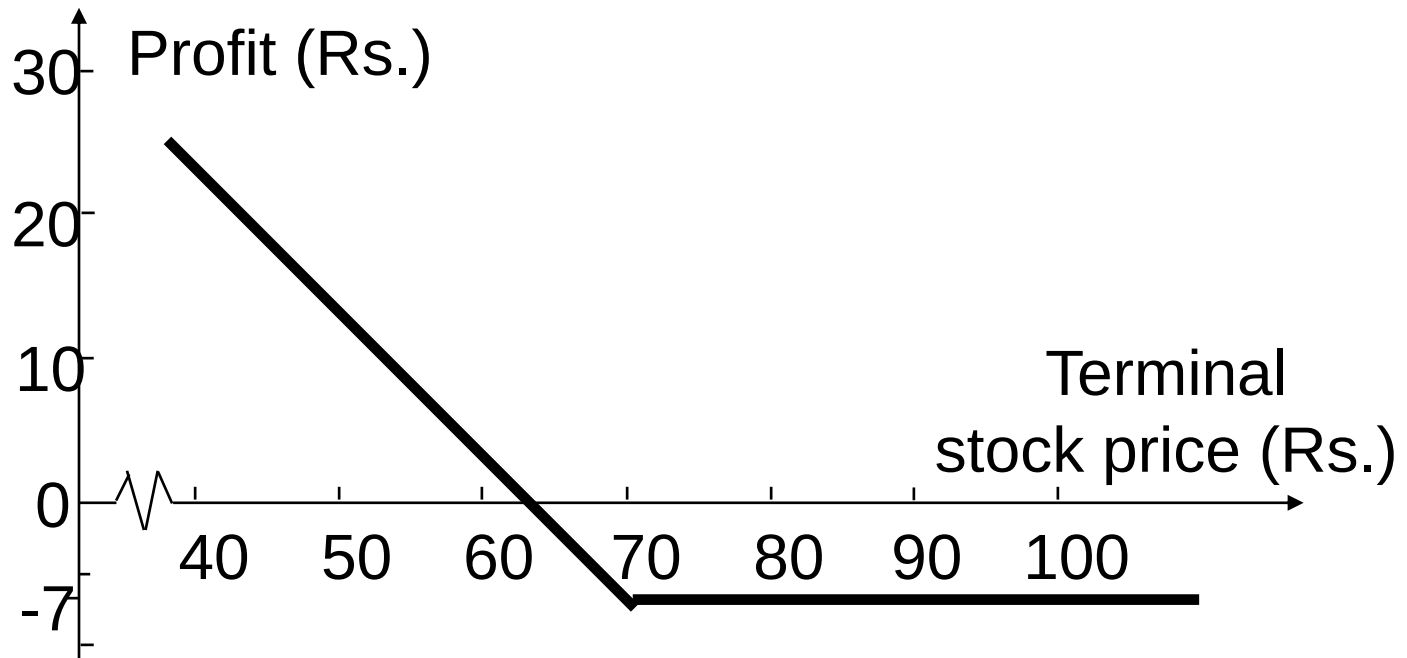
Short Call (Call Writer)

Profit from writing one European call option: option price = Rs. 5, strike price = Rs.100



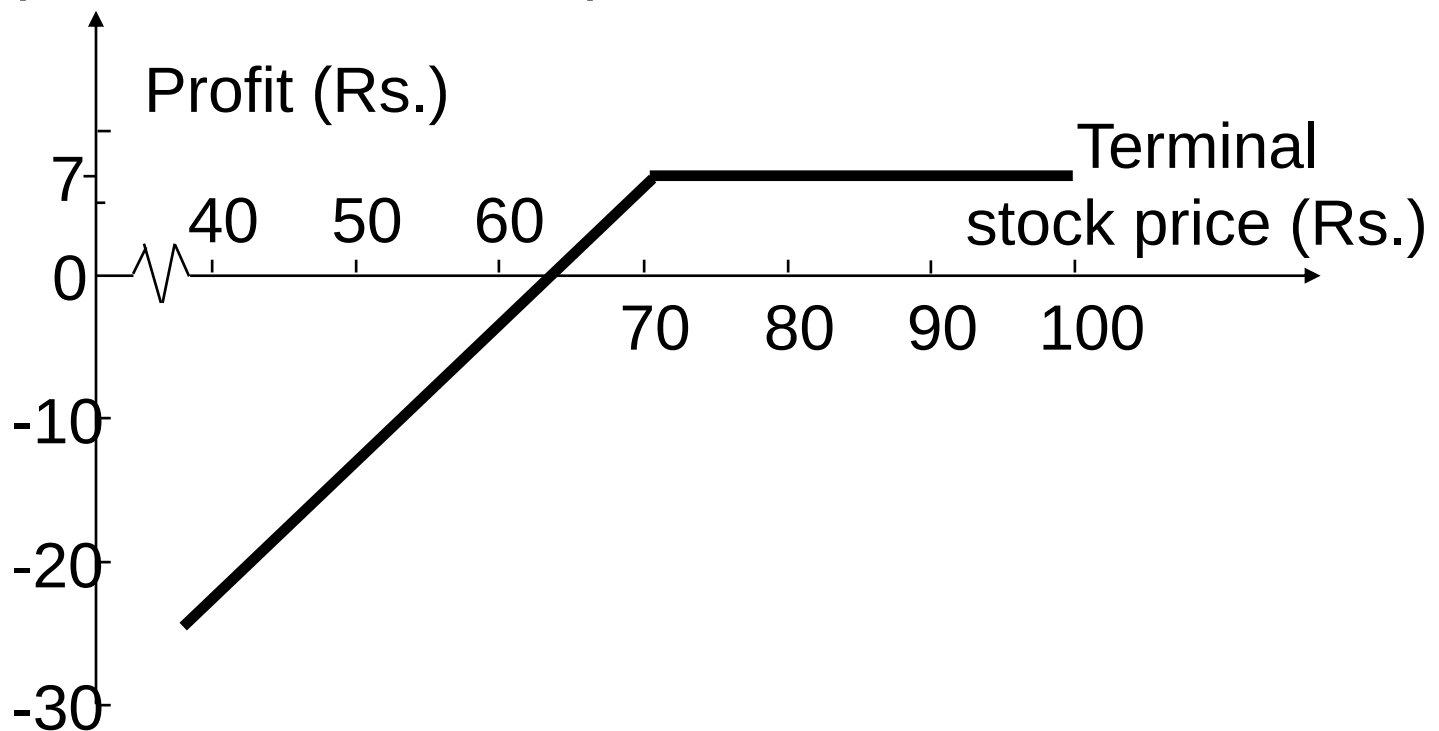
Long Put (Put Holder)

Profit from buying a European put option: option price = Rs.7, strike price = Rs. 70



Short Put (Put Writer)

Profit from writing a European put option: option price = Rs. 7, strike price = Rs. 70

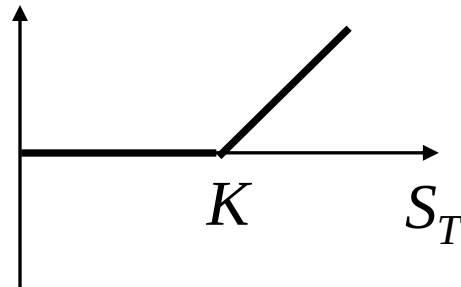


Payoffs from Options

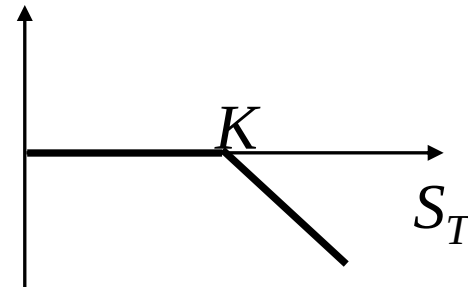
What is the Option Position in Each Case?

K = Strike price, S_T = Price of asset at maturity

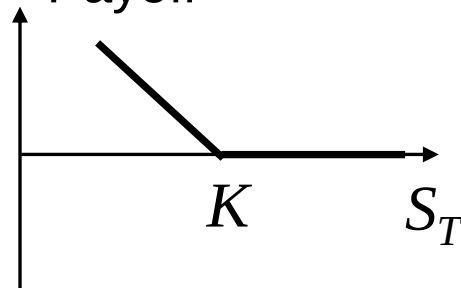
Payoff



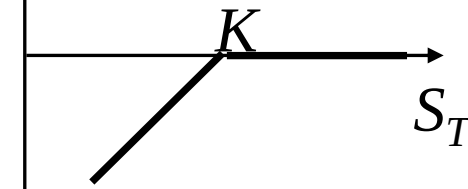
Payoff



Payoff



Payoff



Options vs Futures/Forwards

- A futures/forward contract gives the holder the obligation to buy or sell at a certain price
- An option gives the holder the right to buy or sell at a certain price

Types of Traders

- Trade on price movements (Hedgers)
- Contain risks and uncertainties (Speculators)
- Profit from short term mispricing (Arbitrageurs)

Hedging

- One of the most important and practical applications of Futures and Options is 'Hedging'. In the event of any adverse market movements, hedging is a simple work around to protect your trading positions from making a loss.

Example

- Imagine you have bought 250 shares of Infosys at Rs.2,284/- per share. The quarterly results are expected soon. You are worried Infosys may announce a not so favorable set of numbers, as a result of which the stock price may decline considerably. To avoid making a loss in the spot market you decide to hedge the position.

In order to hedge the position in spot, we simply have to enter a counter position in the futures market. Since the position in the spot is 'long', we have to 'short' in the futures market.

Future price = Rs.2,285/-

Lot size= 250

Infosys Price	Long Spot P&L	Short Futures P&L	Net P&L
2200	$2200 - 2284 = -84$	$2285 - 2200 = +85$	$-84 + 85 = +1$
2290	$2290 - 2284 = +6$	$2285 - 2290 = -5$	$+6 - 5 = +1$
2500	$2500 - 2284 = +216$	$2285 - 2500 = -215$	$+216 - 215 = +1$

Hedging using options

Assume that the following information is given for a put option on Infosys stock.

Option price = Rs.1/-

Strike price= Rs.2,286/-

Lot size= 250

Infosys Price	Long Spot P&L	Put option P&L	Net P&L
2200	$2200 - 2284 = -84$	$2285 - 2200 = +85$	$-84 + 85 = +1$
2290	$2290 - 2284 = +6$	-1	$+6 - 1 = +5$
2500	$2500 - 2284 = +216$	-1	$+216 - 1 = +215$

Speculators

- A speculator is an individual or firm that, as the name suggests, speculates – or guesses – that the price of securities will go up or down and trades the securities based on their speculation.
- Example: You would like to speculate on a rise in the price of a certain stock. The current stock price is \$29 and a 3-month call with a strike price of \$30 costs \$2.90. You have \$5,800 to invest. Identify two alternative investment strategies, one in the stock and the other in an option on the stock. What are the potential gains and losses from each?

Strategy 1: Buy 200 shares from market for \$29 each

Strategy 2: Buy 2,000 call options from derivative market for option price \$2.90

At time “T” the following scenario may occur:

Scenario 1: Underlying stock price becomes \$40

Gain from spot market: $200 \times (\$40 - \$29) = \$2,200$

Gain from option market: $[2,000 \times (\$40 - \$30)] - \$5,800 = \$14,200$

Scenario 2: If the share price goes down to \$25, the first strategy leads to a loss of $200 \times (\$29 - \$25) = \$800$, whereas the second strategy leads to a loss of the whole \$5,800 investment. This example shows that options contain built in leverage.

Arbitrage

- Arbitrage is the strategy of taking advantage of price differences in different markets for the same asset. For it to take place, there must be a situation of at least two equivalent assets with differing prices.
- In essence, arbitrage is a situation where a trader can profit from the imbalance of asset prices in different markets. The simplest form of arbitrage is purchasing an asset in the market where the price is lower and simultaneously selling the asset in the market where the asset's price is higher.



Example

- Explain the arbitrage opportunity when the price of a dually listed mining company stock is \$50 (USD) on the New York Stock Exchange and \$60 (CAD) on the Toronto Stock Exchange. Assume that the exchange rate is such that 1 USD equals 1.21 CAD. Explain what is likely to happen to prices as traders take advantage of this opportunity.

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- In this case, traders can buy shares on the TSX and sell them on the NYSE to lock in a USD profit of $50-60/1.21=0.41$ per share.
 - As they do this the NYSE price will fall and the TSX price will rise so that the arbitrage opportunity disappears

Example (Arbitrage)

- The price of gold is currently \$1,200 per ounce. The forward price for delivery in 1 year is \$1,300 per ounce. An arbitrageur can borrow money at 3% per annum. What should the arbitrageur do? Assume that the cost of storing gold is zero and that gold provides no income.

- The arbitrageur should borrow money to buy a certain number of ounces of gold today and short forward contracts on the same number of ounces of gold for delivery in one year. This means that gold is purchased for \$1,200 per ounce and sold for \$1,300 per ounce. Interest on the borrowed funds will be $0.03 \times \$1,200$ or \$36 per ounce. A profit of \$64 per ounce will therefore be made.

Arbitrage

Arbitrage involves simultaneous buying and selling of a stock in spot and future in order to gain from a difference in the price.

NEAR NEXT FAR

Company	Future	Spot	Basis	Basis %	Previous Basis	Change	Lot Size
MRF	58,560.20	58,425.40	134.80	0.23	230.15	-95.35	10
BOSCHLTD	12,946.00	12,856.35	89.65	0.70	15.55	74.10	50
SHREECEM	20,101.65	20,025.30	76.35	0.38	70.90	5.45	50
COFORGE	1,991.05	1,919.10	71.95	3.75	11.10	60.85	375
PAGEIND	18,560.00	18,503.55	56.45	0.31	50.10	6.35	30
MARUTI	7,227.45	7,190.10	37.35	0.52	23.80	13.55	100
SRF	4,224.20	4,202.05	22.15	0.53	12.95	9.20	250
HEROMOTOCO	2,935.50	2,915.20	20.30	0.70	10.95	9.35	300
NAUKRI	3,353.45	3,337.90	15.55	0.47	11.25	4.30	250
BAJAJFINSV	6,292.05	6,277.20	14.85	0.24	31.10	-16.25	125

Note: Basis is the difference between futures and spot price. Future prices are 5-minute snapshot prices.

Practice Question

- A stock price is \$29. An investor buys one call option contract on the stock with a strike price of \$30 and sells a call option contract on the stock with a strike price of \$32.50. The market prices of the options are \$2.75 and \$1.50, respectively. The options have the same maturity date. Describe the investor's position.

Swaps (OTC)

- Swap is a commitment between two parties to do a series of transactions in the future
- Similar to forward contract, but involves multiple future transactions
- Swaps can involve a variety of underlyings
 - Interest rates
 - Equity
 - Currency
 - Commodities

Derivative Market in India

- Exchange traded derivatives were permitted in the year 2000,
- In the last two decades the derivative market has grown rapidly.
- The equity derivative turnover to GDP ratio for the year 2017-18 - 985%.
- National Stock Exchange of India (NSE) ranked as the second largest stock exchange worldwide in terms of the number of contracts traded.

Dangers

- Traders can switch from being hedgers to speculators or from being arbitrageurs to speculators
- It is important to set up controls to ensure that trades are using derivatives in for their intended purpose



Thank You